

1. What are Python Data Types?

Data types define the type of value that a variable can store.

Category	Data Types
Numeric	int, float, complex
Sequence	str, list, tuple
Set	set, frozenset
Mapping	dict
Boolean	bool
None Type	NoneType

2. What is Dynamic Typing?

Python is a **dynamically typed language**, which means **you do not need to declare the data type of a variable**. Python automatically determines the data type based on the value assigned.

Example

```
x = 10    # int
```

```
x = "Python" # str
```

```
x = 3.14    # float
```

The same variable can store different data types.

3. What is Strong Typing?

Python is a strongly typed language, which means it does not automatically perform invalid operations between incompatible data types. If you try to combine incompatible types, Python raises an error.

Example

```
x = 10
```

```
y = "20"
```

```
print(x + y)
```

Output:

TypeError

To perform the operation, you must convert the data type.

```
print(x + int(y))
```

Output:

30

Strong typing means Python does not allow operations between incompatible data types without explicit conversion.

4. What is the Difference Between Mutable and Immutable?

Mutable objects can be changed after they are created, whereas **immutable objects** cannot be changed once they are created. If we modify an immutable object, Python creates a new object instead of changing the existing one.

Mutable Data Types:

- List
- Set
- Dictionary

Immutable Data Types:

- Integer
- Float
- String
- Tuple
- Boolean

5. What is the Difference Between Ordered and Unordered Data Types?

Ordered data types store elements in the order they are inserted, so they can be accessed using indexes. **Unordered data types** do not guarantee the order of elements, so indexing is not supported.

Ordered Data Types:

- List
- Tuple
- String
- Dictionary

Unordered Data Types:

- Set
- Frozenset